

# BS-PH101 — Physics-I

Semester I/II • 3L:1T:0P • 4 Credits

## Module 2 Comprehensive Exam Notes: Optics

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## 1. Interference vs. Diffraction and Classification

### 1.1 Distinction Between Interference and Diffraction

#### Definition: Interference and Diffraction

- **Interference:** Modification in intensity resulting from the superposition of two or more coherent light waves originating from distinct coherent sources.
- **Diffraction:** The bending of light waves around the sharp edges of an obstacle or aperture and their subsequent superposition within the geometrical shadow region.

#### Comprehensive Comparison:

| Property     | Interference                                                   | Diffraction                                                                                         |
|--------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Wave Origin  | Superposition of waves from two distinct coherent sources.     | Superposition of secondary wavelets originating from different points of the <i>same</i> wavefront. |
| Fringe Width | Fringes are usually of equal width ( $\beta = \lambda D/d$ ).  | Fringes are never of equal width (central max is twice as wide as secondary max).                   |
| Intensity    | All bright fringes have equal intensity ( $I_{\max} = 4I_0$ ). | Intensity decreases rapidly with increasing order of secondary maxima.                              |
| Darkness     | Minima are completely dark ( $I_{\min} = 0$ ).                 | Minima are generally not completely dark.                                                           |

### 1.2 Fraunhofer vs. Fresnel Diffraction

| Feature   | Fresnel Diffraction                                      | Fraunhofer Diffraction                                   |
|-----------|----------------------------------------------------------|----------------------------------------------------------|
| Distance  | Source and screen are at finite distances from aperture. | Source and screen are at effectively infinite distances. |
| Wavefront | Incident wavefront is spherical or cylindrical.          | Incident wavefront is plane (parallel rays).             |
| Lenses    | No lenses required to focus pattern.                     | Convex lenses required before and after aperture.        |
| Analysis  | Analyzed using Fresnel half-period zones.                | Analyzed using simple wave phase summation.              |

## 2. Fraunhofer Diffraction at Single, Double, and Multiple Slits

### 2.1 Single Slit Diffraction

Consider a plane wavefront of wavelength  $\lambda$  incident normally on a slit of width  $e$ .

**Single Slit Intensity Formula**

The resultant intensity on a screen at diffraction angle  $\theta$  is:

$$I(\theta) = I_0 \left( \frac{\sin \beta}{\beta} \right)^2 \quad \text{where } \beta = \frac{\pi e \sin \theta}{\lambda} \quad (1)$$

**Key Conditions:**

- **Central Maximum:** At  $\theta = 0 \Rightarrow \beta = 0, \lim_{\beta \rightarrow 0} \frac{\sin \beta}{\beta} = 1 \Rightarrow I = I_0$ .
- **Minima Condition:**  $\sin \beta = 0$  (for  $\beta \neq 0$ )  $\Rightarrow \beta = m\pi \Rightarrow e \sin \theta = m\lambda$  ( $m = \pm 1, \pm 2, \pm 3, \dots$ ).
- **Secondary Maxima:** Occur near  $\beta = \pm 1.43\pi, \pm 2.46\pi, \dots$  with intensity ratios:

$$I_0 : I_1 : I_2 = 1 : \frac{1}{22} : \frac{1}{61} = 1 : 0.045 : 0.016 \quad (2)$$

- **Width of Central Maximum:**  $\Delta\theta = \frac{2\lambda}{e} \Rightarrow$  Linear Width  $W = \frac{2\lambda f}{e}$  (where  $f$  is lens focal length).

**2.2 Double Slit Diffraction**

For two parallel slits, each of width  $e$  and separated by opaque spacing  $d$  (grating element  $= e + d$ ):

**Double Slit Intensity Formula**

$$I(\theta) = 4I_0 \left( \frac{\sin \beta}{\beta} \right)^2 \cos^2 \alpha \quad \text{where } \beta = \frac{\pi e \sin \theta}{\lambda}, \quad \alpha = \frac{\pi(e + d) \sin \theta}{\lambda} \quad (3)$$

It is the product of single-slit diffraction factor  $\left( \frac{\sin \beta}{\beta} \right)^2$  and interference factor  $\cos^2 \alpha$ .

**Missing Orders (Absent Spectra):** Occurs when an interference maximum coincides with a diffraction minimum:

$$(e + d) \sin \theta = n\lambda \quad \text{and} \quad e \sin \theta = m\lambda \Rightarrow \frac{e + d}{e} = \frac{n}{m} \quad (4)$$

If  $e = d$ , then  $\frac{n}{m} = 2 \Rightarrow n = 2m$ , so 2<sup>nd</sup>, 4<sup>th</sup>, 6<sup>th</sup> interference orders disappear.

**2.3 Diffraction Grating and Resolving Power**

A plane transmission diffraction grating consists of  $N$  parallel, equally spaced slits with grating element  $(e + d)$ .

**Grating Equations and Resolving Power**

- **Principal Maxima Condition:**

$$(e + d) \sin \theta = m\lambda \quad (m = 0, 1, 2, \dots) \quad (5)$$

- **Dispersive Power ( $D$ ):** Rate of change of diffraction angle with wavelength:

$$D = \frac{d\theta}{d\lambda} = \frac{m}{(e + d) \cos \theta} \quad (6)$$

- **Resolving Power ( $R$ ):** Capacity to separate two spectral lines of wavelength  $\lambda$  and  $\lambda + d\lambda$ :

$$R = \frac{\lambda}{d\lambda} = mN \quad (N = \text{total illuminated rulings}) \quad (7)$$

#### Solved Problem 1: Grating Spectral Lines and Maximum Order

**Problem:** A grating has 5000 lines/cm. Light of wavelength  $\lambda = 6000 \text{ \AA}$  is incident normally. Find: (a) Grating element  $(e + d)$ , (b) Highest observable order.

**Solution:**

1. Grating element  $(e + d) = \frac{1 \text{ cm}}{5000} = 2 \times 10^{-4} \text{ cm} = 2 \times 10^{-6} \text{ m}$ .
2. Maximum angle  $\theta_{\max} = 90^\circ \Rightarrow \sin 90^\circ = 1$ .
3. Substitute in grating equation  $(e + d) \sin 90^\circ = m_{\max} \lambda$ :

$$m_{\max} = \frac{(e + d)}{\lambda} = \frac{2 \times 10^{-6} \text{ m}}{600 \times 10^{-9} \text{ m}} = 3.33 \Rightarrow m_{\max} = 3 \quad (8)$$

Hence, maximum 3<sup>rd</sup> order spectrum can be observed.

#### Solved Problem 2: Single Slit Central Width

**Problem:** Light of wavelength  $\lambda = 5890 \text{ \AA}$  is incident normally on a slit of width 0.1 mm. A convex lens of focal length  $f = 1.0 \text{ m}$  is placed behind the slit. Calculate linear width of central maximum.

**Solution:**

Linear width of central maximum  $W = \frac{2\lambda f}{e}$ .  $W = \frac{2 \times 5890 \times 10^{-10} \text{ m} \times 1.0 \text{ m}}{0.1 \times 10^{-3} \text{ m}} = 11.78 \times 10^{-3} \text{ m} = 11.78 \text{ mm}$ .

## 3. Polarisation of Light

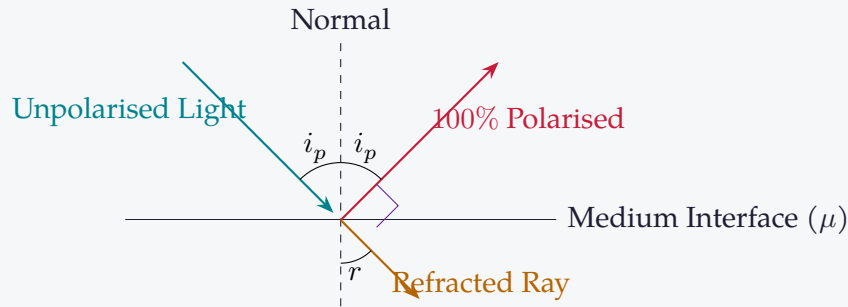
### 3.1 Brewster's Law and Double Refraction

#### Brewster's Law

When unpolarised light is incident at a specific angle  $i_p$  (polarising angle) on a dielectric interface, the reflected light is 100% plane polarised with electric vector perpendicular to plane of incidence.

$$\mu = \tan i_p \quad \text{and} \quad i_p + r = 90^\circ \quad (9)$$

## Polarisation by Reflection (Brewster's Law)



**Double Refraction (Birefringence):** When unpolarised light enters an anisotropic crystal (e.g. Calcite, Quartz), it splits into two plane polarised rays travelling at different speeds:

- **Ordinary Ray (O-ray):** Obeys Snell's law; travels with equal velocity in all directions ( $v_o = c/\mu_o$ ).
- **Extraordinary Ray (E-ray):** Violates Snell's law; velocity depends on direction of propagation ( $v_e$ ).

## 3.2 Retardation Plates: Quarter-Wave and Half-Wave Plates

## Retardation Plate Thickness Equations

For a crystal plate cut parallel to optic axis of thickness  $t$ :

- **Quarter-Wave Plate (QWP):** Introduces phase difference  $\Delta\phi = \pi/2$  or path difference  $\Delta x = \lambda/4$ :

$$t = \frac{\lambda}{4|\mu_o - \mu_e|} \quad (10)$$

- **Half-Wave Plate (HWP):** Introduces phase difference  $\Delta\phi = \pi$  or path difference  $\Delta x = \lambda/2$ :

$$t = \frac{\lambda}{2|\mu_o - \mu_e|} \quad (11)$$

## 3.3 Optical Activity and Laurent's Half-Shade Polarimeter

Optically active substances rotate the plane of polarization of linearly polarized light. The angle of rotation  $\theta$  is given by Biot's law:  $\theta = S \cdot L \cdot c$ .

$$\text{Specific Rotation } [S]_{\lambda}^T = \frac{\theta}{L \cdot c} \quad \left( \text{in } \frac{\text{deg}}{\text{dm} \cdot \text{g/cm}^3} \right) \quad (12)$$

where  $\theta$  = rotation angle (deg),  $L$  = path length in decimeters (dm),  $c$  = concentration in  $\text{g/cm}^3$ .

**Laurent's Half-Shade Polarimeter:** Uses a half-shade plate (half quartz, half glass) to divide the field of view into two halves, enabling precise setting of equal brightness.

**Solved Problem 3: Quarter-Wave Plate Thickness**

**Problem:** Calculate the minimum thickness of a quarter-wave plate of Calcite for sodium light of wavelength  $\lambda = 5893 \text{ \AA}$ . Given  $\mu_o = 1.6583$  and  $\mu_e = 1.4864$ .

**Solution:**

1. Calcite is a negative crystal ( $\mu_o > \mu_e$ ), so  $|\mu_o - \mu_e| = 1.6583 - 1.4864 = 0.1719$ .
2. Formula:  $t = \frac{\lambda}{4(\mu_o - \mu_e)}$ .
3. Substitute:  $t = \frac{5893 \times 10^{-10} \text{ m}}{4 \times 0.1719} = \frac{5893 \times 10^{-10}}{0.6876} = 8570 \times 10^{-10} \text{ m} = \mathbf{0.857 \text{ }\mu\text{m}}$ .

## 4. Lasers (Light Amplification by Stimulated Emission of Radiation)

### 4.1 Principles of Laser Action: Einstein's Coefficients

Interaction of radiation with matter involves three fundamental processes:

#### Three Radiation Processes and Einstein's Relations

1. **Stimulated Absorption:** Atom absorbs photon: Rate  $R_{12} = B_{12}N_1u(\nu)$ .
2. **Spontaneous Emission:** Atom decays naturally: Rate  $R_{21,\text{sp}} = A_{21}N_2$ .
3. **Stimulated Emission:** Incident photon induces decay: Rate  $R_{21,\text{st}} = B_{21}N_2u(\nu)$ .

At thermal equilibrium,  $R_{12} = R_{21,\text{sp}} + R_{21,\text{st}}$ . Equating with Planck's law yields Einstein's relations:

$$B_{12} = B_{21} \quad \text{and} \quad \frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \quad (13)$$

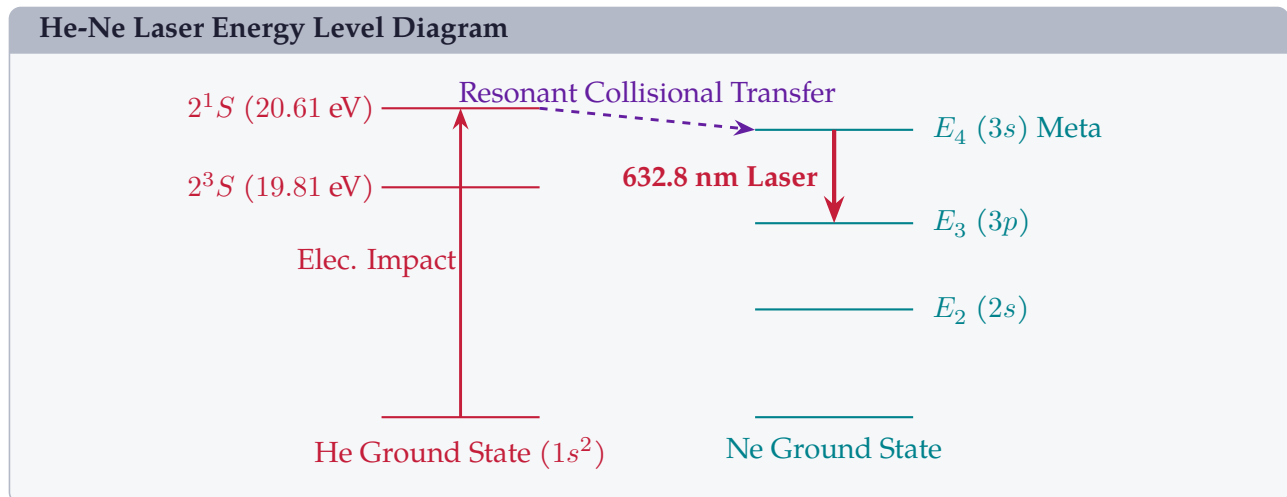
### 4.2 Population Inversion and Pumping

Under normal thermal equilibrium (Boltzmann distribution  $N_2 = N_1 e^{-h\nu/kT}$ ),  $N_1 \gg N_2$ .

#### Definition: Population Inversion and Pumping

- **Population Inversion:** The non-equilibrium state in which higher energy level population exceeds lower energy level ( $N_2 > N_1$ ). Necessary condition for optical amplification.
- **Pumping:** The external process of supplying energy to raise atoms to higher excited levels to achieve population inversion (e.g. Optical pumping, Electrical discharge, Direct electron excitation).
- **Metastable State:** An excited state with a long lifetime ( $\sim 10^{-3} \text{ s}$  compared to normal  $10^{-8} \text{ s}$ ), allowing accumulation of atoms for population inversion.

### 4.3 He-Ne Laser (Gas Laser)



#### Working Mechanism of He-Ne Laser:

1. Electric discharge accelerates electrons which excite He atoms to metastable  $2^1S$  (20.61 eV) state.
2. He atoms transfer energy to Ne atoms via resonant inelastic collisions, populating Ne's  $3s$  ( $E_4$ ) level.
3. Stimulated emission from  $3s \rightarrow 2p$  ( $E_4 \rightarrow E_3$ ) produces red laser light at  $\lambda = 632.8$  nm.

## Quick Revision — Module 2

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### Key Formulas and Metrics at a Glance

|                                |                                                                                                                      |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Single Slit Minima             | $e \sin \theta = m\lambda \quad (m = \pm 1, \pm 2, \dots)$                                                           |
| Single Slit Central Width      | $\Delta\theta = \frac{2\lambda}{e} \implies \text{Linear Width } W = \frac{2\lambda f}{e}$                           |
| Double Slit Missing Orders     | $\frac{e+d}{e} = \frac{n}{m} \quad (n = \text{interf. order}, m = \text{diff. order})$                               |
| Grating Equation               | $(e + d) \sin \theta = m\lambda$                                                                                     |
| Grating Dispersive Power       | $D = \frac{d\theta}{d\lambda} = \frac{m}{(e+d) \cos \theta}$                                                         |
| Grating Resolving Power        | $R = \frac{\lambda}{d\lambda} = mN$                                                                                  |
| Brewster's Law                 | $\mu = \tan i_p \quad \text{and} \quad i_p + r = 90^\circ$                                                           |
| Quarter-Wave Plate (QWP)       | $t = \frac{\lambda}{4 \mu_o - \mu_e }$                                                                               |
| Half-Wave Plate (HWP)          | $t = \frac{\lambda}{2 \mu_o - \mu_e }$                                                                               |
| Specific Rotation              | $[S]_\lambda^T = \frac{\theta}{L \cdot c} \quad (\text{deg} \cdot \text{dm}^{-1} \cdot (\text{g}/\text{cm}^3)^{-1})$ |
| Einstein Coefficients Relation | $\frac{A_{21}}{B_{21}} = \frac{8\pi h \nu^3}{c^3} \quad \text{and} \quad B_{12} = B_{21}$                            |

### One-Line Definitions

- **Diffraction Grating:** An optical component with a periodic structure of  $N$  slits that diffracts light into several beams.
- **Brewster Angle ( $i_p$ ):** Angle of incidence at which light with a particular polarization is perfectly transmitted without reflection.
- **Birefringence:** The optical property of a material having a refractive index that depends on the polarization and propagation direction of light.
- **Population Inversion:** State in which a system exists with more members in an excited state than in lower energy states ( $N_2 > N_1$ ).
- **Metastable State:** An excited atomic state with a long lifetime ( $\sim 10^{-3}$  s), essential for building up population inversion.

### Frequently Asked / Viva Questions

#### Q1. What is the basic difference between interference and diffraction?

Interference results from superposition of waves from two distinct coherent sources; diffraction results from superposition of wavelets from different parts of the same wavefront.

## Frequently Asked / Viva Questions (contd.)

**Q2. Why is central maximum in single-slit diffraction twice as wide as secondary maxima?**

Because central maximum extends between first minima on either side ( $\sin \theta = \pm \lambda/e$ ), spanning angular width  $2\lambda/e$ , whereas secondary maxima span width  $\lambda/e$ .

**Q3. What is meant by missing orders in double slit diffraction?**

Missing orders occur when an interference maximum coincides with a diffraction minimum at the same angle  $\theta$ , making that order absent ( $\frac{e+d}{e} = \frac{n}{m}$ ).

**Q4. State Rayleigh's criterion for spectral resolution.**

Two spectral lines are just resolved when the central principal maximum of one falls on the first minimum of the other.

**Q5. How does resolving power of a grating differ from dispersive power?**

Dispersive power ( $d\theta/d\lambda$ ) measures angular separation between lines; resolving power ( $\lambda/d\lambda = mN$ ) measures ability to distinguish two close wavelengths.

**Q6. State Brewster's law and show reflected and refracted rays are perpendicular.**

$\mu = \tan i_p$ . Since  $\frac{\sin i_p}{\sin r} = \tan i_p = \frac{\sin i_p}{\cos i_p} \Rightarrow \sin r = \cos i_p = \sin(90^\circ - i_p) \Rightarrow i_p + r = 90^\circ$ .

**Q7. What are O-ray and E-ray in a birefringent crystal?**

O-ray (Ordinary) obeys Snell's law with constant speed  $v_o$ ; E-ray (Extraordinary) violates Snell's law with directional speed  $v_e$ .

**Q8. Distinguish between positive and negative uniaxial crystals.**

In positive crystals ( $\mu_e > \mu_o$ , e.g. Quartz),  $v_o > v_e$ ; in negative crystals ( $\mu_o > \mu_e$ , e.g. Calcite),  $v_e > v_o$ .

**Q9. Explain the function of a Quarter-Wave Plate (QWP).**

A QWP introduces a path difference of  $\lambda/4$  ( $\pi/2$  phase shift) between O-ray and E-ray, converting linearly polarized light to circularly or elliptically polarized light.

**Q10. What is population inversion and why is it necessary for laser action?**

Population inversion ( $N_2 > N_1$ ) ensures that rate of stimulated emission exceeds rate of stimulated absorption, producing net optical amplification.

**Q11. Why is Helium gas added to Neon gas in a He-Ne laser?**

He atoms are excited by electron impact to metastable states ( $2^1S, 3^3S$ ) and efficiently transfer energy to Ne atoms via resonant collisions, achieving population inversion in Ne.

**Q12. Derive relation between Einstein A and B coefficients.**

Equating thermal equilibrium rate  $B_{12}N_1u(\nu) = A_{21}N_2 + B_{21}N_2u(\nu)$  with Boltzmann distribution  $N_2/N_1 = e^{-h\nu/kT}$  and Planck's radiation law gives  $B_{12} = B_{21}$  and  $A_{21}/B_{21} = 8\pi h\nu^3/c^3$ .

## Summary Comparison of Laser Processes

| Process                     | Incident Condition                                  | Transition                           | Output Properties                                     |
|-----------------------------|-----------------------------------------------------|--------------------------------------|-------------------------------------------------------|
| <b>Absorption</b>           | Photon<br>( $h\nu = E_2 - E_1$ )                    | Lower $E_1 \rightarrow$ Higher $E_2$ | Atom reaches excited state.                           |
| <b>Spontaneous Emission</b> | Natural decay<br>(lifetime $\sim 10^{-8}\text{s}$ ) | Higher $E_2 \rightarrow$ Lower $E_1$ | Incoherent, random phase/direction.                   |
| <b>Stimulated Emission</b>  | Triggering photon<br>( $h\nu$ )                     | Higher $E_2 \rightarrow$ Lower $E_1$ | Coherent, identical phase, direction, and wavelength. |